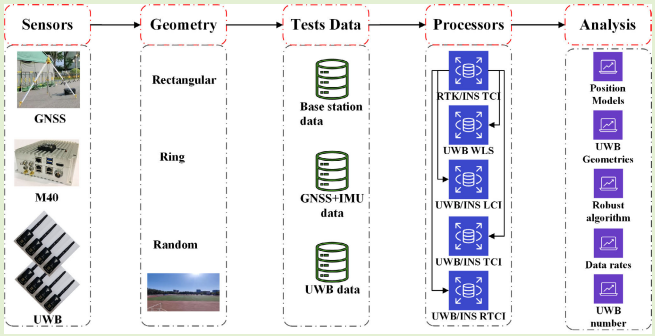
Comprehensive Evaluation of Robust and Tight Integration of UWB and Low-Cost IMU

Lin Chen, Zhouzheng Gao, *Member, IEEE*, Qiaozhuang Xu, Cheng Yang, *Member, IEEE*, and You Li, *Member, IEEE*

***Abstract*—With the rapid development of the Internet of Things (IoT), indoor positioning method with characters of high accuracy, continuity, and reliability has become an urgent requirement. Although the global satellite navigation system (GNSS) can provide that information in open-sky conditions, it still cannot work in GNSS-denied environments. To counteract the drawbacks of GNSS and conventional low-accuracy indoor technologies, this article presents a credible indoor positioning system with integrating ultrawide-band (UWB) measurements and low-cost inertial navigation system (INS) tightly. Based on such a system, the impacts of non-line-of-sight (NLOS) errors, the distribution of UWB base stations, the number of UWB base stations, and the rate of UWB data on UWB/INS tight integration are evaluated comprehensively by analyzing three sets of experimental data and several groups of simulated data. Results demonstrate that: 1) UWB/INS integration upgrades UWB positioning accuracy significantly, with decimeter-level positioning accuracy in the horizontal component and meter-level accuracy in the vertical direction while using the UWB/INS tight integration model; 2) the influences of the geometry structure of the UWB base station on the performance of UWB and the UWB/INS loose integration are obvious, and such an influence on the performance of the UWB/INS tight integration is slight; 3) the positioning accuracy of UWB/INS tight integration increases gradually along with the rate of UWB data; and 4) the presented method shows visible repression of the influence of NLOS errors on UWB/INS tight integration.**



***Index Terms*—Inertial navigation system (INS), non-line-of-sight (NLOS), ultrawide-band (UWB), UWB/INS tight integration.**

I. Introduction

THE development of the Internet of Things (IoT) is changing users' requirements for positioning technologies. Usually, the global satellite navigation system (GNSS) is used for outdoor positioning services. However, GNSS is not suitable for indoor positioning services because of satellite signal blocking [1]. Indoor positioning methods, such as Wi-Fi, Bluetooth low energy (BLE), ZigBee, and radio frequency identification (RFID) are adopted widely [2]. However, the complex indoor environments make it hard for users to receive clean signals. Usually, indoor positioning measurements based on those sensors would be polluted by non-line-of-sight (NLOS) errors. Moreover, the NLOS errors even would block the signal transmission, which will lead to discontinuous solutions. Fortunately, the low-cost motion-sensors-based dead-reckoning (DR) can provide continuous indoor localization solutions [3]. Motion sensors, such as inertial measurement units (IMUs), can measure the acceleration and angular rate of a carrier. These measurements will not be affected by users' environments. However, the solutions of an IMU-based DR would diverge significantly over time. Therefore, the existing indoor methods are hard to provide users with high accuracy and high continuity positioning solutions.

In recent years, ultrawide-band (UWB), which is usually based on the angle of arrival (AOA), time of arrival (TOA), or time difference of arrival (TDOA), has been adopted to provide high-accuracy positioning solutions for indoor

Manuscript received 16 July 2023; accepted 24 August 2023. Date of publication 1 September 2023; date of current version 31 October 2023. This work was supported in part by the National Key Research and Development Program of China under Grant 2021YFB3901301 and in part by the National Natural Science Foundation of China (NSFC) under Grant 42274022. The associate editor coordinating the review of this article and approving it for publication was Dr. Masood Ur Rehman. *(Corresponding author: Zhouzheng Gao.)*

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Digital Object Identifier 10.1109/JSEN.2023.3309623

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applications [4]. This is mainly due to the UWB's advantages of high positioning accuracy, strong anti-interference capability, high penetration, and low power consumption. However, UWB is also affected by signal occlusion and NLOS ranging. To reduce the impact of NLOS errors, an improved reliable UWB positioning algorithm was proposed using the UWB measurements of AOA and TOA in [5], and an indoor positioning experiment was conducted in a building by using 14 UWB tags to analyze the effect of multipath signals on the positioning results in the building. In [6], the Kalman filter based on iterative and anti-difference algorithms was introduced to the UWB positioning system, which can enhance the stability of UWB positioning. The effect of users' environments on UWB ranging is analyzed in [7]. According to the tests on the propagation data, the excessive propagation delay of the LOS blockage material limits the UWB ranging accuracy. In [8], a single-diameter detection algorithm with less computation and complexity is given to correct the UWB data with low accuracy. In [9], the NLOS error is corrected by using the combination of particle filtering and traceless Kalman filtering, which can limit the NLOS error significantly. Besides, the antenna phase center offset calibration and the multipoint time latency determination are proposed in [10], which are to reduce the impact of system errors on UWB positioning accuracy.

To overcome those drawbacks in UWB positioning, UWB is to integrate with the inertial navigation system (INS). At present, the integration between UWB and INS can be divided into loose integration and tight integration [11]. The UWB/INS loose integration is based on UWB positioning results and that of INS, and the UWB/INS tight integration is realized by integrating UWB measurements and the INS-predicted ranges [12]. The precondition of the loose integration is that the number of the available UWB ranges should not be less than three. Otherwise, the loose integration mode will be retrograded into INS-only mode [13]. For example, [14] adopted the extended Kalman filter to integrate multipoint UWB localization and IMU data. In [15], the particle filter was utilized in the UWB/INS integration system, which is to solve the tracking loss problem in pedestrian tracking. The works in [16] concluded that the positioning accuracy of the UWB/INS integration outperforms the two independent subsystems. The identification of nonvisual conditions was used in [17] to upgrade the position accuracy of the UWB/INS loose integration. The results showed that the accuracy of the horizontal localization could be better than 1.5 m. Generally, most of the works at present focus on the horizontal positioning performance of UWB-related technologies. In [18], the UWB/INS loose integration system enhanced by non-holonomic constraint and Rauch–Tung–Striebel (RTS) smoother was proposed. The results illustrate that it can provide centimeter-level position, millimeter-level velocity, and less than 0.15° attitude, respectively. The results based on 30-s serious NLOS simulations demonstrate that 3-D positioning accuracy with an accuracy better than 20 cm still can be obtained by using such an enhanced UWB/INS integration system. In [19], the EKF is adopted to tightly integrate the inertial sensors with the ranges of the UWB. The algorithm was tested by simulation data and only provided 2-D localization. A tightly coupled UWB/INS system for indoor walking applications is proposed in [20], which confirms the superiority of the tight integration. The stability and accuracy of the UWB/INS integration could be furtherly improved by applying the radial basis function-based neural network [21]. A convolutional neural network (CNN)-based UWB/INS positioning algorithm in [22] can provide a centimeter positioning accuracy.

Besides, researchers focused on the impact of UWB anchor geometry structure on positioning accuracy [23]. Remarkably, the accuracy, availability, and reliability of positioning accuracy would be affected significantly by the geometry structure of UWB anchors. Usually, the position dilution of precision (PDOP) is adopted to analyze the geometry structure, and it is a direct index for UWB positioning accuracy [24]. Therefore, the optimal base station distribution problem becomes the problem to get a smaller PDOP. Methods like the genetic algorithm [25] and particle swarm optimization algorithm [26] were used to find the distribution of UWB base stations that could satisfy the relatively minimum PDOP. In [27], the relationship between the pseudo-satellite distribution and the geometric accuracy factor was analyzed through simulation experiments. In [28], the relationship between target nodes and user tags in 2-D localization was investigated. In [29], indoor space was divided into single and multigroup base station service spaces. In the single group base station service space, PDOP was used to establish the base station distribution model, and a genetic algorithm was used to solve iteratively to find the optimal base station location. A comparative study was performed for the three distributions (Y-shape, L-shape, and mirror-based) in [30], and the cuboid-shape distribution was considered a commonly used one in terms of accuracy and compactness. In [31], four types of UWB base station distribution were simulated, and PDOP was used to assess the effect of the base station distribution on the UWB's positioning accuracy. Experiments have proved that the eight-base-station distribution has the most accurate positioning accuracy than the four-base-station.

The above studies mainly focus on the 2-D UWB positioning, this article carries out a tight integration of the TOA-based UWB ranges and low-cost INS. Then, three sets of GNSS, UWB, and INS data were processed to present the positioning accuracy of different UWB/INS integration methods and the impacts of NLOS errors, the distribution of UWB base station, the number of UWB base station, and the rate of UWB data on UWB/INS integration. Compared to current works on UWB/INS integration, the contributions of this work can be summarized as: 1) we provide a residual-based robust UWB/INS tight integration model to restrain the impacts of the NLOS errors, by which reliable 3-D positioning solutions can be obtained and 2) the impacts of the UWB-based positioning model, the number of UWB anchors, UWB anchors' distribution, and UWB data rate on positioning accuracy are analyzed systematically. The article is arranged as follows. Section II introduces the models of UWB positioning, UWB/INS loose integration, and robust UWB/INS tight integration. Sections III and IV show the discussions and conclusion, respectively.

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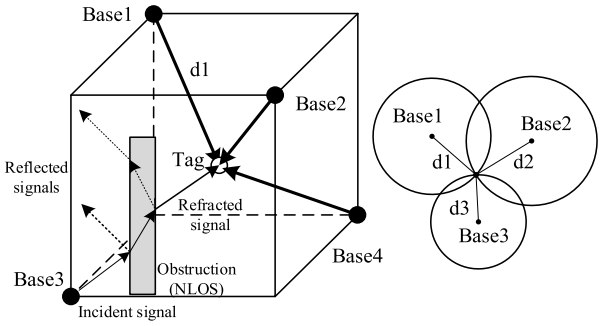


Fig. 1. TOA-based UWB positioning.

II. Methods

A. Mathematical Models of Ultrawide-Band Positioning

The basic principle of the TOA-based localization can be described in Fig. 1. Similar to the GNSS positioning, the UWB measurements are the distances (d_k, $k = 1, 2, \ldots, N$, N is the number of UWB base stations) between tag and UWB base stations. Due to the complexity of the signal transmission environment, the signal cannot always reach the tag directly. This results in NLOS errors. Then, the intersection of these ranges may not be found, which requires an adjustment algorithm to estimate the location of the intersection point. The adjustment methods often used here are least squares (LS) and weighted least squares (WLS).

Assuming that the coordinates of the kth base station and the coordinates of the target are (x_k, y_k, z_k) and (x, y, z), then the distance (d_k) can be expressed by

$$d_k = \sqrt{(x_k - x)^2 + (y_k - y)^2 + (z_k - z)^2} \tag{1}$$

and the WLS can be applied for target coordinate estimation [7], which can be written as

$$\begin{bmatrix} x & y & z \end{bmatrix}^T = \left(A^T P A\right)^{-1} A^T P L \tag{2}$$

where A, P, and L are respectively the designed matrix, weight matrix, and the innovation vector. Wherein, the innovation vector can be obtained by making differentia operations between UWB ranges and the corresponding theoretical value.

B. Model of UWB/INS Robust and Tight Integration

As mentioned above, enough available UWB ranges are the precondition for UWB positioning. However, such a condition would not be satisfied while suffering serious NLOS errors. It will lead to un-continuous UWB positioning. Therefore, the INS is used to bridge the UWB positioning solutions. The UWB/INS integration can be divided into loose integration and tight integration. Usually, the extended Kalman filter is utilized to realize such two integrations. It can be realized after two phases, wherein one is the time update, and the other is the measurement update. The time update can be written by

$$\begin{bmatrix} \mathbf{x}_{k/k-1} \\ \boldsymbol{P}_{k/k-1} \end{bmatrix} = \begin{bmatrix} \boldsymbol{\Phi}_{k/k-1}\mathbf{x}_{k-1} + \mu_{k-1} \\ \boldsymbol{\Phi}_{k/k-1}\boldsymbol{P}_{k-1}\boldsymbol{\Phi}_{k/k-1}^T + \mathbf{Q}_{k-1} \end{bmatrix} \tag{3}$$

and the measurement update can be described by

$$\begin{bmatrix} \mathbf{x}_k \\ \boldsymbol{P}_k \end{bmatrix} = \begin{bmatrix} \mathbf{x}_{k/k-1} + \boldsymbol{K}_k(\boldsymbol{z}_k - \boldsymbol{H}_k\mathbf{x}_{k/k-1}) \\ (\mathbf{I} - \boldsymbol{K}_k\boldsymbol{H}_k)\boldsymbol{P}_{k/k-1}(\mathbf{I} - \boldsymbol{K}_k\boldsymbol{H}_k)^T + \boldsymbol{K}_k\boldsymbol{R}_k\boldsymbol{K}_k^T \end{bmatrix} \tag{4}$$

where $\mathbf{x}_{k-1}$, $\mathbf{x}_{k/k-1}$, and $\mathbf{x}_k$ denote the parameter vector at $k-1$ epoch, the parameter vector at k epoch predicted by $\mathbf{x}_{k-1}$, and the estimated parameter vector at k epoch, and the corresponding variance are $\boldsymbol{P}_{k-1}$, $\boldsymbol{P}_{k/k-1}$, and $\boldsymbol{P}_k$; $\boldsymbol{\Phi}_{k/k-1}$ is the state transition matrix that is to describe the time-variation of parameters from $k-1$ epoch to k epoch; μ_{k-1} stands for the state noise with the corresponding variance of $\mathbf{Q}_{k-1}$; $\boldsymbol{K}_k$ and $\boldsymbol{R}_k$ are the gain matrix of the Kalman filter and the prior variance matrix of observations; and $\boldsymbol{z}_k$ and $\boldsymbol{H}_k$ denote the innovation vector and the designed coefficient matrix respectively [32].

1) State Model: As mentioned above, the state transition matrix is based on the state function of each parameter. For both loose integration and tight integration of UWB/INS, the state parameter vectors are the same, which can be expressed by

$$\mathbf{x}_k = \left[\boldsymbol{\delta p}_{\mathrm{INS}}^n, \boldsymbol{\delta v}_{\mathrm{INS}}^{\mathbf{n}}, \boldsymbol{\delta\psi}, \boldsymbol{\delta b}_{\mathrm{acc}}, \boldsymbol{\delta b}_{\mathrm{gyro}}, \boldsymbol{\delta s}_{\mathrm{acc}}, \boldsymbol{\delta s}_{\mathrm{gyro}}\right]^T \tag{5}$$

where $\boldsymbol{\delta p}_{\mathrm{INS}}^n$, $\boldsymbol{\delta v}_{\mathrm{INS}}^{\mathbf{n}}$, and $\boldsymbol{\delta\psi}$ are the corrections of position, velocity, and attitude, respectively; $\boldsymbol{\delta b}_{\mathrm{gyro}}$ and $\boldsymbol{\delta s}_{\mathrm{gyro}}$ are the bias and scale factor of gyroscopes; $\boldsymbol{\delta b}_{\mathrm{acc}}$ and $\boldsymbol{\delta s}_{\mathrm{acc}}$ are the bias and scale factor of accelerometers.

In this work, the PSI-angle error model [33] and the first Gauss–Markov process are adopted to express the state dynamic function of navigation parameters (position, velocity, and attitude) and the IMU sensor errors (biases and scale factors). The corresponding expressions can be given by

$$\begin{bmatrix} \delta\dot{\boldsymbol{p}}_{\mathrm{INS},k}^n \\ \delta\dot{\boldsymbol{v}}_{\mathrm{INS},k}^n \\ \boldsymbol{\delta\dot{\psi}}_k \\ \boldsymbol{\delta b}_{\mathrm{acc},k} \\ \boldsymbol{\delta b}_{\mathrm{gyro},k} \\ \boldsymbol{\delta s}_{\mathrm{acc},k} \\ \boldsymbol{\delta s}_{\mathrm{gyro},k} \end{bmatrix} = \begin{bmatrix} \boldsymbol{\omega}_{en}^n \times \boldsymbol{\delta p}^n + \boldsymbol{\delta v}^n \\ \boldsymbol{C}_b^n\boldsymbol{f}^b \times \boldsymbol{\psi} + \boldsymbol{C}_b^n\boldsymbol{\delta f}^b - \left(\boldsymbol{\omega}_{ie}^n + \boldsymbol{\omega}_{in}^n\right) \times \boldsymbol{\delta v}^n + \boldsymbol{\delta g}^n \\ -\boldsymbol{\omega}_{in}^n \times \boldsymbol{\psi} - \boldsymbol{C}_b^n\boldsymbol{\delta\omega}_{ib}^b \\ e^{(-\Delta t/T_{b,\mathrm{acc}})} \cdot \mathbf{b}_{\mathrm{acc},k-1} + \boldsymbol{\eta}_{b,\mathrm{acc},k-1} \\ e^{(-\Delta t/T_{b,\mathrm{gyro}})} \cdot \mathbf{b}_{\mathrm{gyro},k-1} + \boldsymbol{\eta}_{b,\mathrm{gyro},k-1} \\ e^{(-\Delta t/T_{s,\mathrm{acc}})} \cdot \mathbf{s}_{\mathrm{acc},k-1} + \boldsymbol{\eta}_{s,\mathrm{acc},k-1} \\ e^{(-\Delta t/T_{s,\mathrm{gyro}})} \cdot \mathbf{s}_{\mathrm{gyro},k-1} + \boldsymbol{\eta}_{s,\mathrm{gyro},k-1} \end{bmatrix} \tag{6}$$

where $\boldsymbol{\omega}_{ib}^b$ and $\boldsymbol{f}^b$ are the angular rate and specific force measured by gyroscopes and accelerometers in the body frame (b, forward-bottom-right); $\boldsymbol{\omega}_{ie}^n$ denotes the angular rate of the Earth-centered-Earth-fixed frame (e) to the inertial frame (i) projected to the navigation frame (n, north-east-down); $\boldsymbol{\omega}_{in}^n$ denotes the angular rate of the n-frame respect to the i-frame projected to the n-frame; $\boldsymbol{\omega}_{en}^n$ is the angular rate of the n-frame respect to the e-frame projected to the n-frame; $\mathbf{g}^n$ and $\boldsymbol{C}_b^n$ are gravity vector in the n-frame and the transformation matrix from the b-frame to the n-frame; and Δt and T are the IMU interval and relative time of IMU sensor error.

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Based on the state models above, the state transition matrix $\boldsymbol{\Phi}_{k/k-1}$ can be obtained [34]

$$\boldsymbol{\Phi}_{k/k-1}=\begin{bmatrix}\boldsymbol{\Phi}_1 & \boldsymbol{\Phi}_2 & 0 & 0 & 0 & 0 & 0\\ \boldsymbol{\Phi}_3 & \boldsymbol{\Phi}_4 & \boldsymbol{\Phi}_5 & \boldsymbol{\Phi}_6 & 0 & \boldsymbol{\Phi}_7 & 0\\ 0 & 0 & \boldsymbol{\Phi}_8 & 0 & \boldsymbol{\Phi}_9 & 0 & \boldsymbol{\Phi}_{10}\\ 0 & 0 & 0 & \boldsymbol{\Phi}_{11} & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & \boldsymbol{\Phi}_{12} & 0 & 0\\ 0 & 0 & 0 & 0 & 0 & \boldsymbol{\Phi}_{13} & 0\\ 0 & 0 & 0 & 0 & 0 & 0 & \boldsymbol{\Phi}_{14}\end{bmatrix} \tag{7}$$

with, as in (8), shown at the bottom of the page, where $\boldsymbol{\Phi}_{12}=\boldsymbol{\Phi}_{13}=\boldsymbol{\Phi}_{14}=\boldsymbol{\Phi}_{11}$; R_N, R_M, and h are the meridional circle radius, prime unitary circle radius, and geodetic altitude, respectively.

Based on the models above, the time-update phase of the Kalman filter can be used.

2) Measurement Model: The observation function can be expressed by

$$z_k=\boldsymbol{H}_k\mathbf{x}_k+V_k,\quad V_k\sim N(0,\boldsymbol{R}_k) \tag{9}$$

where V_k denotes the residual vector [32].

For the UWB/INS loose integration, the innovation vector can be obtained by making a difference between the UWB-calculated solutions and the INS-predicted solutions

$$z_k=\boldsymbol{p}_{\mathrm{INS},k}-\boldsymbol{p}_{\mathrm{UWB},k}+\iota_{\boldsymbol{p}} \tag{10}$$

where $\boldsymbol{p}_{\mathrm{UWB}}$ and $\boldsymbol{p}_{\mathrm{INS}}$ are the position solutions calculated by UWB and INS; $\iota_{\boldsymbol{p}}$ is the lever-arm correction [35] for the position, which is due to the different measuring centers for UWB and IMU, with the expressions of

$$\iota_{\boldsymbol{p}}=C_b^n l^b \tag{11}$$

where l^b represents the lever-arm measured in the b-frame.

Based on the parameter vector and innovation function above, the designed coefficient matrix for loose integration can be obtained by making error disturbance on the innovation vector around the parameter vector

$$\boldsymbol{H}_k=\begin{bmatrix}\mathbf{I}_{3\times3} & 0_{3\times3} & C_{\mathrm{b}}^n\iota^b\times & 0_{3\times12}\end{bmatrix} \tag{12}$$

where symbols have the same meanings as mentioned above.

For tight integration, the innovation vector can be written as

$$z_k=\begin{bmatrix}d_{\mathrm{INS},1}-d_{\mathrm{UWB},1}\\ d_{\mathrm{INS},2}-d_{\mathrm{UWB},2}\\ \cdots\\ d_{\mathrm{INS},N}-d_{\mathrm{UWB},N}\end{bmatrix}+\begin{bmatrix}\iota_d\\ \iota_d\\ \vdots\\ \iota_d\end{bmatrix} \tag{13}$$

with

$$\iota_d=\left|C_b^n l^b\right| \tag{14}$$

where d_{UWB} and d_{INS} are UWB ranges and the INS-predicted values; ι_d denotes the lever-arm correction in terms of range.

Based on the parameter vector and innovation function of tight integration, the designed coefficient matrix for loose integration can be obtained. As the relationship between the innovation function and parameter vector is nonlinear, the innovation needs to be linearized before making an error disturbance. In general, the coefficient matrix can be expressed as

$$\boldsymbol{H}_k=\begin{pmatrix}(A\cdot C_1)_{j\times3} & 0_{j\times3} & (A\cdot C_1\cdot C_2)_{j\times3} & 0_{j\times12}\end{pmatrix} \tag{15}$$

with

$$C_1=\begin{bmatrix}\dfrac{-(R_N+h)\cos\lambda\sin\varphi}{R_M+h} & -\sin\lambda & -\cos\varphi\cos\lambda\\ \dfrac{-(R_N+h)\sin\varphi\sin\lambda}{R_M+h} & \cos\lambda & -\cos\varphi\sin\lambda\\ \dfrac{\left[R_N\left(1-e^2\right)\right]\cos\varphi}{R_M+h} & 0 & -\sin\varphi\end{bmatrix} \tag{16}$$

$$C_2=C_b^n I_{\mathrm{UWB}}^b\times \tag{17}$$

where R_M and R_N represent the radius of curvature of the meridian circle and the radius of curvature of the unitary circle.

Then, the measurement update of the Kalman filter can be applied for parameter estimation.

3) Residual Information-Based Robust Kalman Filter: The UWB measurements theoretically can be classified into clean measurements, slightly polluted measurements, and seriously polluted measurements, respectively. As listed in the

$$\begin{bmatrix}\boldsymbol{\Phi}_1\\ \boldsymbol{\Phi}_2\\ \boldsymbol{\Phi}_3\\ \boldsymbol{\Phi}_4\\ \boldsymbol{\Phi}_5\\ \boldsymbol{\Phi}_6\\ \boldsymbol{\Phi}_7\\ \boldsymbol{\Phi}_8\\ \boldsymbol{\Phi}_9\\ \boldsymbol{\Phi}_{10}\\ \boldsymbol{\Phi}_{11}\end{bmatrix}=\begin{bmatrix}\mathbf{I}-\left(\boldsymbol{\omega}_{en}^n\times\right)\Delta t\\ \mathbf{I}\Delta t\\ -\mathrm{diag}\left((R_M+h)^{-1}\quad (R_N+h)^{-1}\quad 2\left(\sqrt{R_NR_M}+h\right)^{-1}\right)\mathbf{g}\Delta t\\ \mathbf{I}-\left(\left(\boldsymbol{\omega}_{ie}^n+\boldsymbol{\omega}_{in}^n\right)\Delta t\right)\times\\ \left(\boldsymbol{f}^n\times\right)\Delta t\\ \boldsymbol{C}_b^n\Delta t\\ \boldsymbol{C}_b^n\boldsymbol{f}^b\Delta t\\ \mathbf{I}-\left(\boldsymbol{\omega}_{in}^n\times\right)\Delta t\\ -\boldsymbol{C}_b^n\Delta t\\ \boldsymbol{C}_b^n\boldsymbol{\omega}_{ib}^b\Delta t\\ e^{-\Delta t/T}\end{bmatrix} \tag{8}$$

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measurement-update phase of the Kalman filter, the element in the prior variance R_k will determine the effect of each UWB measurement on the parameter estimation. If the slightly polluted measurements and seriously polluted measurements are set with the same prior variance values as that of the clean measurements, the estimated parameters would be degraded. Therefore, to restrain the impacts of the unclean UWB measurements on parameter estimation, robust factors obtained from the IGG-II model [36] are employed in our work

$$\gamma_k = \begin{cases} 1, & |\widetilde{v}_k| \le c_0 \\ \dfrac{|\widetilde{v}_k|}{c_0} \times \left(\dfrac{c_1 - c_0}{c_1 - |\widetilde{v}_k|}\right), & c_0 < |\widetilde{v}_k| \le c_1 \\ \infty, & |\widetilde{v}_k| > c_1 \end{cases} \tag{18}$$

where c_0 (setting to 1.0–1.5) and c_1 (setting to 2.5–8.0) are two constants; $\widetilde{v}_k$ can be written as

$$\widetilde{v}_k = \frac{V_k}{\sqrt{\boldsymbol{P}_k}} = \frac{\boldsymbol{z}_k - \boldsymbol{H}_k \mathbf{x}_k}{\sqrt{\left(\boldsymbol{H}_k \boldsymbol{P}_{k,k-1} \boldsymbol{H}_k^T + \boldsymbol{R}_k\right)_{i,i}}}. \tag{19}$$

Then, the equivalent variance matrix can be rewritten as

$$\overline{\boldsymbol{R}}_k = \begin{pmatrix} \gamma_1 \boldsymbol{R}_{(1,1)} & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \gamma_N \boldsymbol{R}_{(N,N)} \end{pmatrix} \tag{20}$$

where γ_k is the robust factor of the covariance for the kth measurement.

Then, based on the equivalent variance, the Kalman gain matrix ($\bar{\boldsymbol{K}}_k$) can be recalculated by

$$\bar{\boldsymbol{K}}_k = \boldsymbol{P}_{k,k-1} \boldsymbol{H}_k^T \left(\boldsymbol{H}_k \boldsymbol{P}_{k/k-1} \boldsymbol{H}_k^T + \bar{\boldsymbol{R}}_k\right)^{-1} \tag{21}$$

and the robust parameter vector can be reestimated by

$$\begin{bmatrix} \bar{\mathbf{x}}_k \\ \bar{\boldsymbol{P}}_k \end{bmatrix} = \begin{bmatrix} \mathbf{x}_{k/k-1} + \bar{\boldsymbol{K}}_k (\boldsymbol{z}_k - \boldsymbol{H}_k \mathbf{x}_{k/k-1}) \\ (\mathbf{I} - \bar{\boldsymbol{K}}_k \boldsymbol{H}_k) \boldsymbol{P}_{k/k-1} (\mathbf{I} - \bar{\boldsymbol{K}}_k \boldsymbol{H}_k)^T + \bar{\boldsymbol{K}}_k \boldsymbol{R}_k \bar{\boldsymbol{K}}_k^T \end{bmatrix} \tag{22}$$

where $\bar{\mathbf{x}}_k$ and $\bar{\boldsymbol{P}}_k$ are the robust estimated parameter vector and the corresponding variance matrix.

For such UWB-based positioning systems, the critical initial information that should be given are initial position, velocity, and attitude. For the initial position in WLS, the Bancroft algorithm [37] is applied to provide the initial value of tag coordinates.

III. Tests and Evaluations

A. Tests Scenes and Data Collection Platform

To validate the performance of the UWB/INS tight integration system and the effects of different base station deployment schemes on positioning accuracy, three field tests were arranged in the playground of the China University of Geosciences (Beijing), with eight UWB base stations (marked as A0, A1, A2, A3, A4, A5, A6, and A7) distributing in a rectangular shape (98 m × 108 m), ring shape (103 m × 178 m), and random distribution (98 m × 158 m) (as shown in Fig. 2). The coordinates of these base stations are listed in Table I. Visibly, they are little differences in the vertical coordinates, which will lead to a rather low accuracy in vertical.

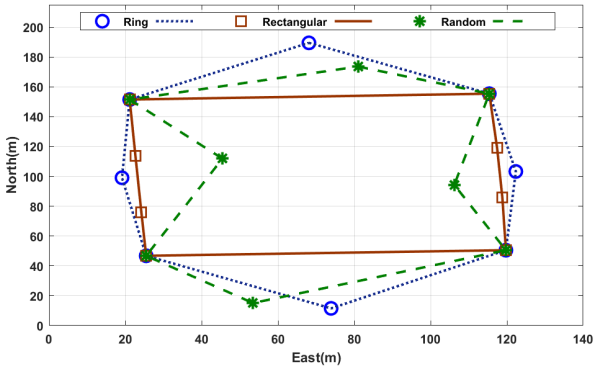


Fig. 2. Three distributions (ring, rectangular, and random) of the eight base stations.

TABLE I
Coordinates of UWB Base Stations in the Three Distributions

Items	UWB base station coordinates		
	Rectangular	Ring	Random
A0	(117.42, 119.26, 2.06)	(68.12, 189.49, 2.11)	(81.04, 173.58, 2.05)
A1	(115.39, 155.41, 2.15)	(115.39, 155.41, 2.15)	(115.39, 155.41, 2.15)
A2	(118.80, 86.04, 2.10)	(122.34, 103.28,2.14)	(106.21, 94.30, 2.07)
A3	(119.74, 50.52, 2.15)	(119.74, 50.52, 2.15)	(119.74, 50.52, 2.15)
A4	(24.21, 75.95, 2.06)	(73.94, 11.53, 2.07)	(53.47, 15.10, 2.08)
A5	(25.47, 46.76, 2.13)	(25.47, 46.76, 2.13)	(25.47, 46.76, 2.13)
A6	(22.65, 113.87, 2.11)	(19.14, 99.06, 2.16)	(45.46, 111.88, 2.11)
A7	(21.15, 151.49, 2.17)	(21.15, 151.49, 2.17)	(21.15, 151.49, 2.17)



Fig. 3. GNSS/INS/UWB integration test.

The equipment and observing environments are depicted in Fig. 3. The playground is almost an open-sky environment, which allows continuous tracking of GNSS. Then, the solution calculated by the RTK/INS tight integration can be used as reference values for the evaluation of the UWB-based solutions. However, for the investigation of our robust method as described in (18)–(22), experimental members course part of these eight UWB base stations in order to simulate the NLOS polluted-low-accuracy UWB measurements.

In this test, the UWB sensors are Nooploop LinkTrack P (https://www.nooploop.com/en/) with the functions of both communication and ranging. Meanwhile, this LinkTrack P has finished the time synchronization between UWB and IMU, which makes it possible to integrate UWB and INS directly. According to the specifications provided by Nooploop Company, the maximum ranging distance and the recommended ranging distance are ≤500 and ≤300 m, which can completely cover the three distributions of a base station.

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TABLE II
TYPICAL SPECIFICATIONS OF LINK TRACK UWB SENSOR

Parameters	Values
Antenna type	External antenna
Maximum range coverage	≤500 m
Recommended range coverage	≤300 m
Update Rate(Hz)	20
Delay(ms)	<50

TABLE III
DETAILS OF LOW-COST IMU SENSOR

Parameters	Values
Gyroscope bias ($°/h$)	10
Accelerometer bias(mGal)	1500
Angular random walk ($°/\sqrt{h}$)	0.35
Velocity random walk ($m/s/\sqrt{h}$)	0.20

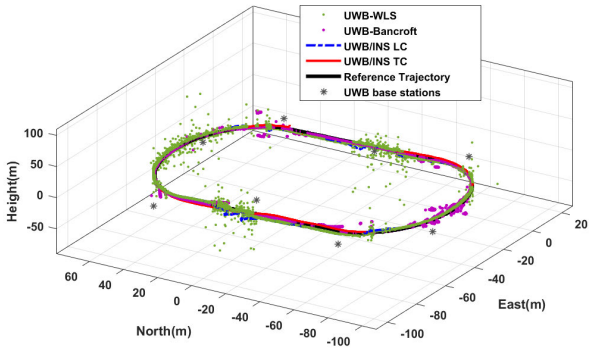


Fig. 4. Test trajectories of different positioning modes based on the random distribution of the UWB base stations.

During the test, the UWB tag and the MEMS IMU (M40) were fixed on the experimental cart with a data rate of 20 and 200 Hz, respectively. The basic information of the UWB and IMU are listed in Tables II and III. The coordinates of each UWB base station in the local coordinate system were measured using a total station, and then these coordinates were projected into the WGS-84 coordinate system. The collected data, including GNSS, UWB, and IMU, were processed in five modes, namely UWB positioning with WLS algorithm (marked as UWB-WLS), UWB positioning with Bancroft algorithm (marked as UWB-Bancroft), the UWB/INS loose integration (marked as UWB/INS LC), the UWB/INS tight integration (marked as UWB/INS TC), and the robust EKF-based UWB/INS tight integration (marked as robust UWB/INS TC), respectively. Herein, the results of the smoothed RTK/INS tight integration were used as the reference values for the performance evaluation of these UWB-based modes.

B. Geometry Structures and Quality of the Three Tests

Fig. 4 shows the trajectories calculated by the UWB-WLS, the UWB-Bancroft, the UWB/INS LC, the UWB/INS TC, and RTK/INS tight integration, respectively, with the eight base stations distributed in random shape. It can be seen that the trajectory of the UWB-WLS and UWB-Bancroft deviate from the reference path in several periods. In contrast, the trajectories of the other modes are close to the reference one. The corresponding HDOP and VDOP are shown in Fig. 5, and the HDOP and VDOP are 1.0 and 12.6 on average, from which it can be seen that the HDOP is much smaller than that of the VDOP. It would lead to a higher positioning accuracy in horizontal components. Meanwhile, the residuals of each data processing mode are calculated and presented in Fig. 6, where visible and frequent big errors can be found. This could result in lower grade positioning accuracy than the theoretical ones.

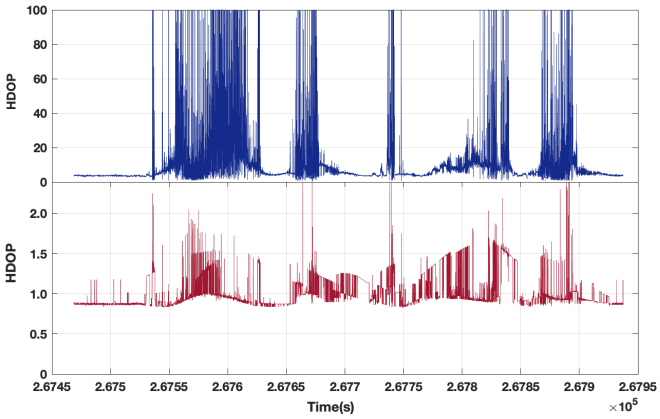


Fig. 5. HDOP and VDOP during the test.

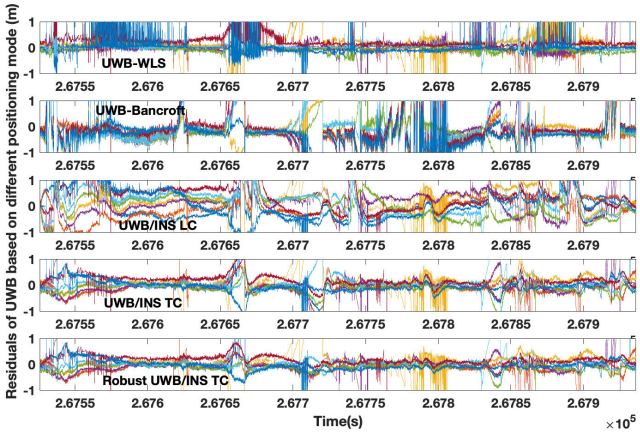


Fig. 6. Residuals of UWB data calculated by the UWB-WLS.

C. Impact of the Robust Algorithm on UWB/INS Tight Integration

Significantly as shown in Fig. 7, the UWB/INS tight integration based on robust EKF performs much better than the other four methods. According to the statistics in Table IV, the position root mean square (rms) values of the UWB/INS tight integration are 40.2, 34.1, and 268.7 cm in the north, east, and vertical components, which are more accurate than that of the UWB/INS loose integration (67.9, 60.3, and 354.9 cm), Bancroft algorithm (79.2, 152.8, and 342.0 cm), and WLS algorithm (142.5, 114.2, and 850.1 cm). In contrast, UWB/INS

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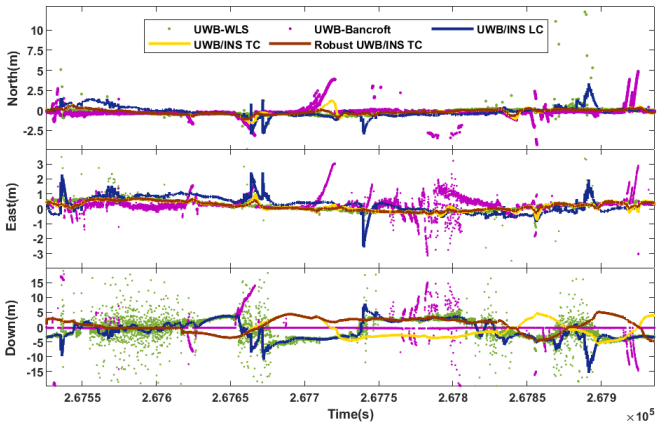


Fig. 7. Positioning errors calculated by UWB, UWB/INS loose integration, and UWB/INS tight integration with and without robust algorithm.

TABLE IV
RMS OF POSITION USING DIFFERENT DATA PROCESSING METHODS

Processing Models	Position (m)		
	North	East	Down
UWB-WLS	1.425	1.142	8.501
UWB-Bancroft	0.792	1.528	3.420
UWB/INS LC	0.679	0.603	3.549
UWB/INS TC	0.402	0.341	2.687
Robust UWB/INS TC	0.338	0.329	2.503

tight integration provides more than 71.81%, 70.12%, and 68.38% improvements compared to the WLS algorithm in the three components. Such improvements are 49.24%, 77.68%, and 21.43% while using the Bancroft algorithm. Compared to the UWB/INS loose integration, the tight integration still can provide more than 40.82%, 43.44%, and 24.28% improvements. Besides, it is significant the UWB positioning accuracies in both horizontal and vertical components from these tests are low than the theoretical ones. This is because of NLOS errors. The NLOS errors have been presented in Fig. 6 and analyzed in advance. Their impacts on UWB ranges are presented in Fig. 15, which will degrade the accuracy in parameter estimation. Besides, as shown in Fig. 8, the cumulative distribution function (cdf) of positioning error in different data processing methods proves that the UWB/INS tight integration based on the robust EKF performs much higher than the other two methods.

D. Impacts of Base Station Distributions on UWB/INS Tight Integration

During the experiment, the HDOP and VDOP with the eight base stations under the rectangular distribution, ring distribution, and random distribution were analyzed, and the results are shown in Figs. 9–14. It can be concluded that the variation trend of HDOP is different from that of VDOP while a tag moves within the area designated by the eight base stations. Generally, the VDOP values of the three types of UWB base station distributions are larger than that of HDOP. It is because HDOP and VDOP are determined separately by

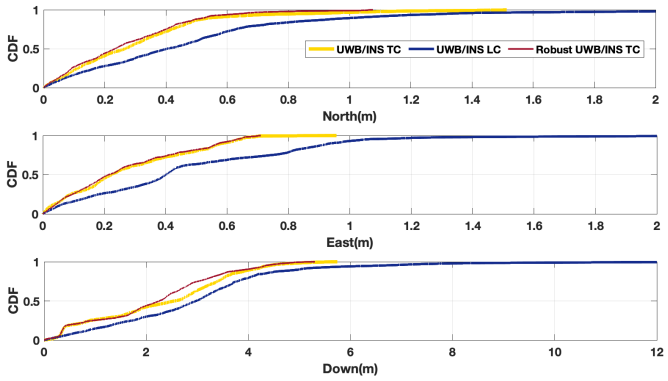


Fig. 8. Cumulative distribution function (cdf) of positioning error.

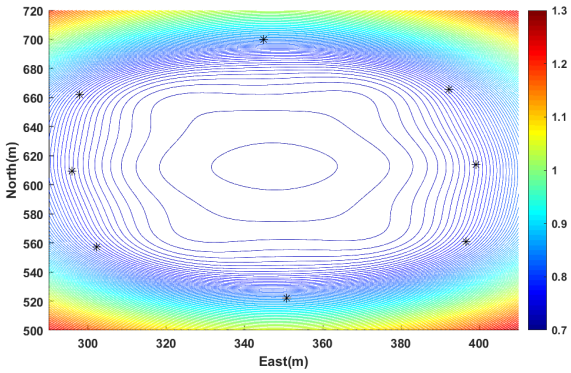


Fig. 9. HDOP distributions of the ring distribution.

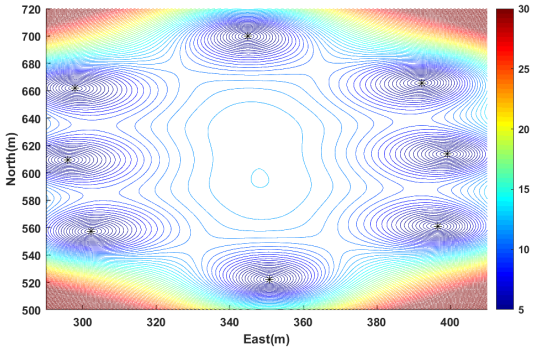


Fig. 10. VDOP distributions of the ring distribution.

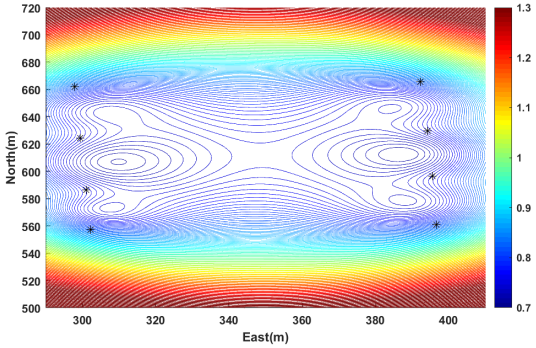


Fig. 11. HDOP distributions of the rectangular distribution.

the geometry structure strength between a tag and all base stations in the horizontal component and vertical direction, respectively. Since the distribution of these base stations in

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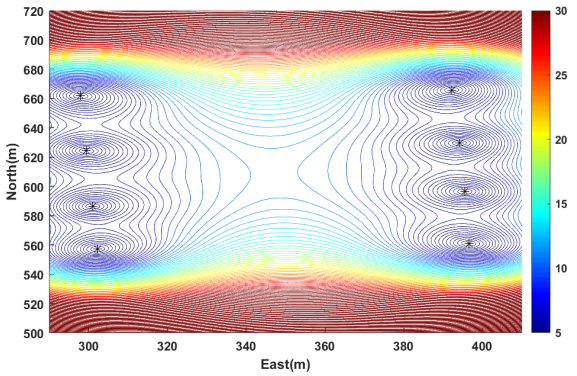


Fig. 12. VDOP distributions of the rectangular distribution.

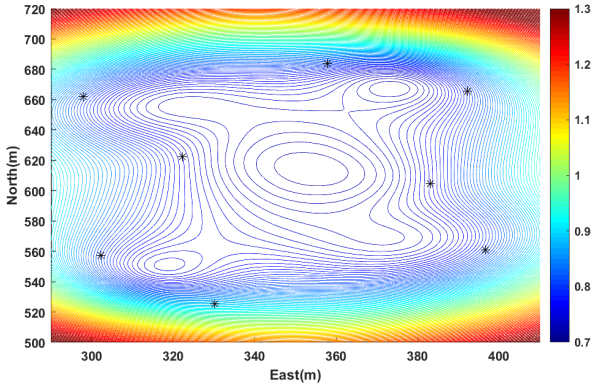


Fig. 13. HDOP distributions of the random distribution.

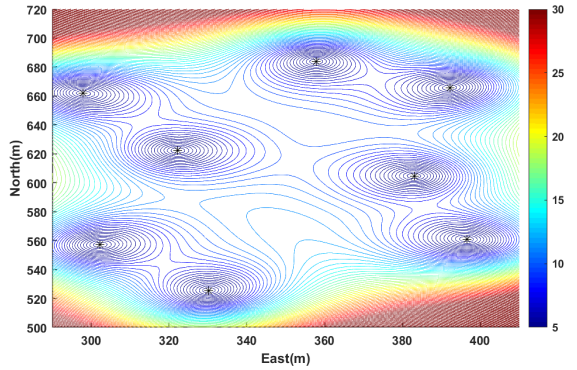


Fig. 14. VDOP distributions of the random distribution.

horizontal component is rather uniform and the base stations in vertical are almost in a plane. Therefore, the geometry structure strength in horizontal is much stronger than that in vertical, which leads to the HDOP being much smaller than VDOP. Besides, the variation trends for HDOP and VDOP are also different. As shown, the values of PDOP become smaller while the tag moves to the center of the shape formed from these base stations. While the VDOP value presents a completely opposite appearance in that VDOP is smaller when the tag gets close to each base station. This appearance is the most direct consequence that the values of HDOP and VDOP changed along with the geometry structure strength. When the tag moves outsides the base station's designated area, both HDOP and VDOP become larger. In contrast, the ring shape of the base station distribution (see Figs. 9 and 10) presents

TABLE V
DOP VALUES OF DIFFERENT BASE STATION DISTRIBUTIONS

Base Station distribution	HDOP		VDOP	
	Inside	Outside	Inside	Outside
Ring	0.7531	0.8727	10.0434	14.1776
Rectangular	0.7607	1.0885	9.0086	27.6186
Random	0.7752	0.9638	9.2723	17.0555

TABLE VI
RMS OF POSITION OF DIFFERENT BASE STATION DISTRIBUTIONS

Processing Models	Position (m)		
	North	East	Down
UWB-only (Ring)	0.464	1.074	4.772
UWB-only (Random)	1.426	1.142	8.502
UWB-only (Rectangular)	0.909	1.835	10.549
UWB/INS LC (Ring)	0.454	0.451	3.259
UWB/INS LC (Random)	0.679	0.603	3.549
UWB/INS LC (Rectangular)	0.601	0.999	3.965
Robust UWB/INS TC (Ring)	0.391	0.438	3.054
Robust UWB/INS TC (Random)	0.338	0.329	2.503
Robust UWB/INS TC (Rectangular)	0.421	0.414	2.541

the best HDOP and VDOP values, and it is followed by the random distribution (see Figs. 11 and 12) and the rectangular distribution (see Figs. 13 and 14). Besides, the average values of HDOP and VDOP of these three distributions in Table V, from which we can find a significant difference under the tag situations of inside or outside of the base stations covered area.

These UWB/INS data are processed using the UWB-only mode, the UWB/INS loose integration mode, and UWB/INS tight integration based on robust EKF mode. According to the statistics in Table VI, the UWB positioning rms values based on the ring shape distribution are 46.4, 107.4, and 477.2 cm in the north, east, and vertical components, which is the highest one. Compared to the solutions based on the random distribution, about 67.42%, 5.94%, and 43.87% in the three components are obtained. Similarly, such improvements are 38.94%, 41.46%, and 54.77% compared to rectangular distribution.

For the UWB/INS loose integration, it is similar to the appearance of UWB only. The ring distribution also provided the best results, which is due to the reason that the positioning accuracy of the UWB/INS loose integration depends highly on the position accuracy of the UWB. Such a phenomenon satisfies the conclusions output from the DOP analysis. As mentioned above, the HDOP and VDOP of the ring distribution are significantly better than those of the other two schemes. However, such an appearance does not exist while using the UWB/INS tight integration based on robust EKF. According to the statistics, it can be known that the random distribution instead of ring distribution provides the optimal solution among the three schemes while using the UWB/INS tight integration based on robust EKF. This is because the UWB/INS tight integration is based on the UWB ranges,

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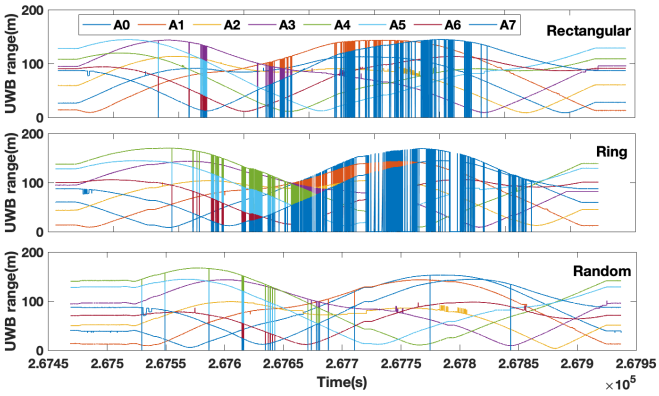


Fig. 15. Measurements of each UWB tag-base-station pair under different base station distributions.

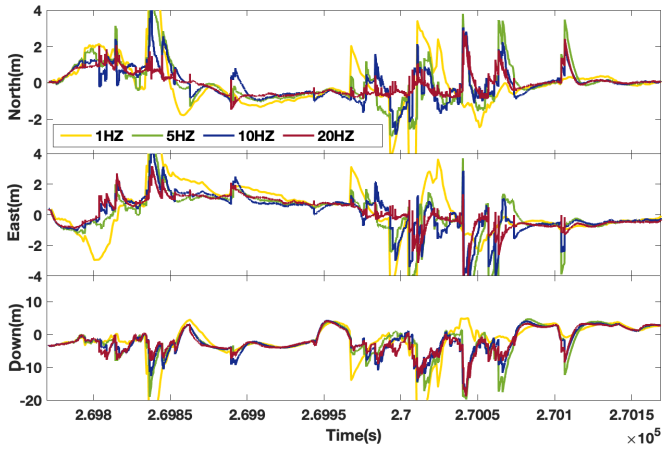


Fig. 16. Position differences of the UWB/INS loose integration by using four types of UWB data (20, 10, 5, and 1 Hz).

which has a greater correlation with the ranging quality. As shown in Fig. 15, there are some visible low-accuracy ranges under the three deployment scenarios, and the ranges in the ring and rectangular distributions are influenced seriously. As shown in Table VI, the positioning accuracy of the UWB/INS tight integration based on robust EKF under the random distribution is the best. Experiments have proved that the positioning accuracy of UWB only and the UWB/INS loose integration could be further optimized by setting reasonable base station distribution. However, the performance of the UWB/INS tight integration majorly depends on UWB-ranging quality. The influence of base station distribution on the performance of the UWB/INS tight integration was much weaker compared to the other two methods.

E. Impacts of UWB Data Rates on UWB/INS Tight Integration

To evaluate the effect of UWB sampling rate on UWB/INS loose integration and tight integration based on robust EKF results, the original UWB data were resampled to 10, 5, and 1 Hz, respectively. The loose integration and tight integration based on robust EKF solutions were calculated. It is shown in Table VII that the positioning accuracy gradually decreases while the UWB sampling rate decreases.

TABLE VII
POSITION RMS WHILE USING DIFFERENT RATE UWB DATA

Methods	Position (m)		
	North	East	Down
UWB/INS LC (20hz)	0.601	0.999	3.965
UWB/INS LC (10hz)	0.843	1.181	4.195
UWB/INS LC (5hz)	1.117	1.495	4.751
UWB/INS LC (1hz)	1.700	2.428	6.943
Robust UWB/INS TC (20hz)	0.421	0.415	2.541
Robust UWB/INS TC (10hz)	0.448	0.426	2.616
Robust UWB/INS TC (5hz)	0.478	0.433	2.648
Robust UWB/INS TC (1hz)	0.609	0.456	3.491

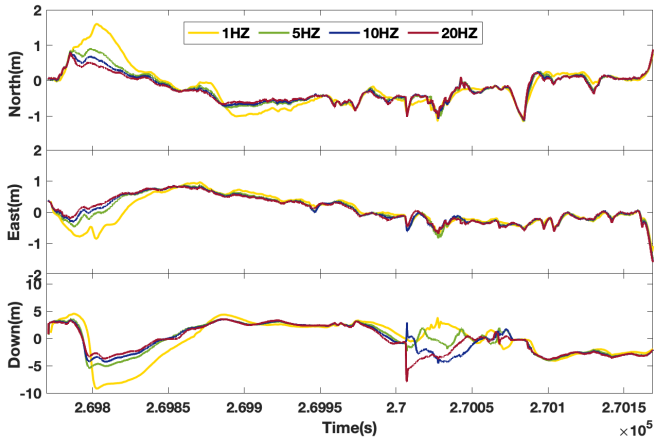


Fig. 17. Position differences of the UWB/INS tight integration based on robust EKF by using four types of UWB data (20, 10, 5, and 1 Hz).

The position differences of the UWB/INS loose integration and tight integration based on robust EKF calculated by the four types of UWB data (20, 10, 5, and 1 Hz) are shown in Figs. 16 and 17. The position rms of UWB/INS loose integration based on UWB rate of 20 Hz are 60.1, 99.9, and 396.5 cm in the north, east, and vertical directions. Compared to those based on 1 Hz UWB data, the position accuracy reductions are 64.67%, 58.82%, and 42.89% in the three directions. Compared to the UWB/INS tight integration based on 1 Hz UWB data, the position rms of the UWB/INS tight integration based on 20 Hz UWB data provides about 32.65%, 9.00%, and 27.24% improvements in the three directions. It can be concluded that the UWB sampling rate plays a significant impact on the UWB/INS integration, and such an effect on the loose integration is more significant than that on the tight integration.

F. Impacts of Base Station Number on UWB/INS Tight Integration

The density of the UWB base stations would affect the performance of UWB [38]. The PDOP value is related to the number of observable UWB base stations. Theoretically, the value of PDOP becomes smaller, and the positioning accuracy will be higher along with the increasing number of UWB base stations [39]. In this part, the effects of base station numbers on the performance of the UWB/INS tight integration are presented. The UWB data is divided into three groups, wherein group one includes eight base stations, group two includes six

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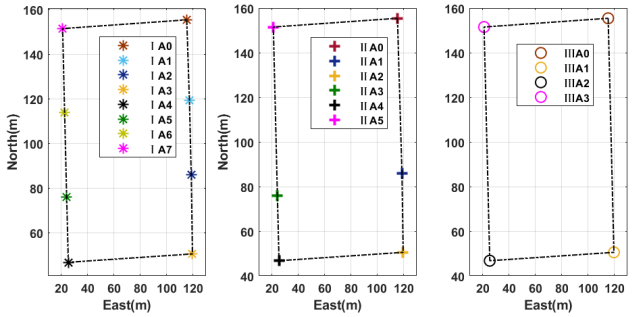


Fig. 18. Distributions of eight base stations, six base stations, and four base stations.

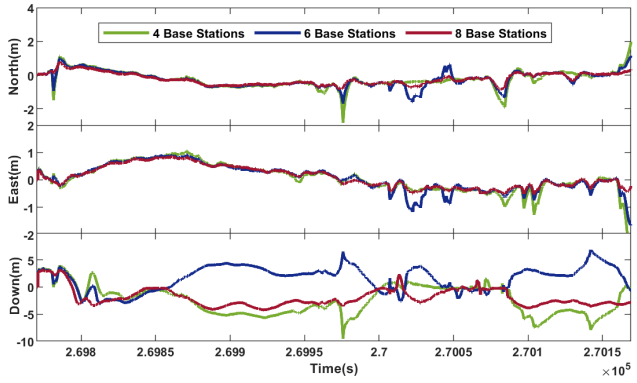


Fig. 19. Position errors based on eight base stations, six base stations, and four base stations.

TABLE VIII
RMS VALUES OF POSITION WHILE USING DIFFERENT NUMBER OF BASE STATIONS

Processing Models	Position (m)		
	North	East	Down
Robust UWB/INS TC (8 base stations)	0.421	0.414	2.541
Robust UWB/INS TC (6 base stations)	0.507	0.484	2.756
Robust UWB/INS TC (4 base stations)	0.544	0.558	3.571

base stations, and group three contains four base stations (as shown in Fig. 18).

The three groups' data are processed by the UWB/INS tight integration based on robust EKF. The position errors calculated based on the three groups' UWB data are presented in Fig. 19. It can be seen that the number of available UWB base stations affects the position accuracy of UWB/INS tight integration visibly. From Table VIII, the position rms based on four UWB base stations are 54.4, 55.8, and 357.1 cm in the north, east, and vertical directions. Compared to the solutions based on six base stations, the reductions in position accuracy are 20.50%, 14.32%, and 7.81% in the three directions. Compared to the solutions based on eight base stations, the reductions in position accuracy are 22.57%, 25.77%, and 28.85% in the three directions, respectively. It can be seen that the number of UWB base stations performs a significant impact on the positioning of the UWB/INS tight integration. It is shown in Table VIII that more available UWB base stations provide higher accuracy of the position of the UWB/INS tight integration.

G. Experiment Discussion and Summarization

According to the results outcome from parts of Sections III-C to III-F, it is visible that the positioning accuracy of the UWB/INS tight integration in NLOS condition could be upgraded significantly after applying the IGG-II robust theory. A similar conclusion can also be found in [40], in which the robust method was used to identify observations with NLOS errors and mitigate their effects on UWB/INS localization. Besides, as pointed out in [41] and [42], the factor graph model and neural networks could also be helpful methods to restrain the impacts of NLOS signals in the UWB-based positioning methods. Our tests also demonstrated that the geometry distributions of the UWB base station would affect UWB-related positioning accuracy directly in terms of HDOP and VDOP and the ring distribution could be recommended according to our tests. However, such a conclusion may be negated while suffering NLOS errors. To avoid the impacts of the geometry distributions of the UWB base station, the gray wolf optimizer algorithm [43] could be a helpful method. Besides, the number of UWB base stations and the rate of UWB data for the same geometry distribution also determined the positioning accuracy of UWB/INS tight integration. In general, we provided comprehensive evaluations on the impacts of NLOS errors, the geometry distributions of UWB base stations, the number of UWB base stations, and UWB data rate on the positioning accuracy of UWB/INS integration, and all of these factors should be considered carefully while utilizing the UWB-based positioning methods.

IV. CONCLUSION

To satisfy the demand for seamless high-accuracy positioning indoors and outdoors and to solve the limited application of GNSS in the indoor environment, this article builds the positioning method to combine the UWB measurements and INS tightly. To evaluate the performance of the presented method, practical experiments are carried out to verify and analyze the influences of UWB base station distributions, UWB base station numbers, and UWB data interval on positioning accuracy. The results illustrate that the positioning accuracy of the UWB/INS tight integration is higher than that of UWB-only and UWB/INS loose integration. Meanwhile, more base stations would increase the possibility of providing higher positioning accuracy. A similar conclusion can also be obtained while using different interval UWB data. However, the performance of the UWB/INS tight integration is slightly affected by the distribution of the UWB base stations, which means that it is more convenient to apply UWB/INS tight integration in practical applications compared to UWB-only and UWB/INS loose integration.

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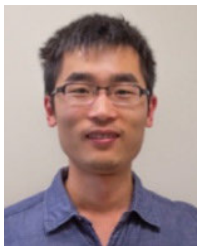
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